

Target architecture, CRM flow and implementation plan

With developer responsibilities for a team of four

Item	Detail
Status	Architecture final. All twelve open decisions settled. Codebase audited on 25 September 2026. Ready to start Week 0.
Scope	A working prototype of the Exporter CRM: companies, qualification, sales conversations, deals and paperwork, background checks, and handovers to other teams.
Team	Four developers working at the same time, each with their own area and files.
Duration	About five to six weeks including a three-day Week 0 (indicative, see section 7).
This document	Self-contained. It replaces the earlier Implementation Plan v1.1 wherever the two differ. Nothing in it depends on earlier conversations or files.

How to read it

- **Managers and stakeholders:** section 1 (summary), section 3 figures, section 4 (the flow), section 6 (decisions) and section 7 (plan).
- **Developers:** everything, and in particular your own part of section 9, plus the shared rules in sections 7 and 8. File paths and audit evidence are in Appendix A.
- **Unfamiliar terms** are explained in Appendix B. The full architecture poster is the last page (zoom in; it is a vector page).

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1. Summary

What we are building. An internal CRM that staff use to find exporting companies, decide which are worth pursuing, manage the sales relationship, gather paperwork for each financing deal, record background checks, and hand finished work to the teams that decide on lending and keep the money records. The exporter never logs in; everything is done by staff.

Why it is changing. Today the CRM pushes every company along one long line of ten statuses, from LEAD to ACTIVE. That line mixes different questions: *is this company a fit?*, *how is the conversation going?*, *is it safe to work with them?* and *is there a deal?* A company can only sit in one place, so it cannot be, for example, a promising prospect whose background check is flagged. The CEO asked for a design that separates these questions.

The new design. Each company has one short journey, LEAD → PROSPECT → CUSTOMER, plus three gauges that move on their own: **qualification**, **conversation** and **background check**. Deals are separate records, many per company, each with its own buyer and paperwork. A problem with one deal or buyer never marks the customer as bad. Lending decisions, money records and the internal workings of compliance checks stay outside the CRM; the CRM hands over to them.

Where we stand. The current CRM works and is well tested for what it does, but none of the gauges and no deal record exist yet. The biggest hidden job is that a company has no name of its own in the CRM: its name lives in an old table we must stop depending on. Three known security problems are fixed and several remain. Tests pass apart from 27 known failures unrelated to the CRM.

The plan. A three-day Week 0 splits the four most crowded files and fixes the shared agreements (contracts) between developers. Then four developers work in parallel for about five weeks: Developer 1 on safety and shared foundations, Developer 2 on the company record and qualification (the critical path), Developer 3 on conversations, deals and paperwork, and Developer 4 on background checks and handovers. Because we start fresh with sample data, there is no data migration.

Before anyone outside the team uses the prototype: all of Developer 1's security fixes and Developer 4's clean-up of misleading placeholders must be in. **Before any real exporter data or documents:** a real virus scanner and secure cloud storage are also required (section 7.6).

2. Background and context

2.1 What the system is

An internal CRM for exporters combined with a verification and compliance record. Staff create companies, record contacts and activity, track the relationship, run and record checks, and hand work on. Companies come from two kinds of source:

- **RXIL**, a partner that sends exporters it has already filtered, together with its own results (KYC, AML/CFT, invoice duplication, vessel tracking, e-bill of lading, buyer rating, insurance). We store RXIL's results exactly as given and **never recompute them**.
- **Everyone else**: companies we find ourselves (research, databases, web searches, manual entry, bulk import) or who approach us directly. For these we do qualification and checks ourselves.

Technology: Vite/React frontend, FastAPI backend, Postgres database. CRM code lives under `backend/app/modules/onboarding` and `frontend/src/modules/onboarding`.

2.2 Why the redesign

The CEO's view, in short: the CRM should manage *discovery, qualification, engagement and the customer lifecycle*. Qualification criteria are filters, not stages. Contacting happens after a company becomes a prospect. A company that says "not now" is a future customer, not a dead lead. A red flag is not necessarily permanent (a legitimate exporter can briefly appear on a sanctions list because of politics). And customer quality and transaction quality are different things: one bad transaction out of a hundred does not make a bad customer. Work should start manual and become automated later, so every decision must record whether a person or a computer made it.

2.3 Scope of the CRM

The CRM does	The CRM does not do
Collect leads and maintain the company database	Decide whether to lend (underwriting and credit)
Apply qualification criteria and keep the reasons	Keep money records (the ledger)
Turn qualified leads into prospects	Run the internal logic of compliance checks
Manage the sales conversation, including "not now" follow-ups	Process transactions or track how they perform
Identify real deals and gather their paperwork	Complex data science or automated scoring (later)
Record background-check outcomes on the company	Exporter self-service login
Maintain customer status, and hand over to other teams	Anything on the legacy onboarding workflow (2.4)

2.4 The two onboarding paths

The codebase contains two onboarding paths. The older one (the `onboarding_request` table, a Temporal workflow and an 18-state machine) is **formally out of scope**: its worker is not started and its switch is off. **Never build on it**. The live path is the Exporter CRM (`exporter_profile`, contacts, activities, the `verification-results` table and the screening checklist), and this is what we are reshaping. One complication: the company's name and country are still stored in the legacy table today, so Developer 2 moves them into the CRM (task L2-03).

2.5 The module rule, and its one exception

All connecting code lives in the onboarding module. **No file in the `kyb`, `cases`, `customers` or `compliance` modules is edited**. Onboarding may use those modules; nothing may use onboarding. Reasons: those modules were delivered against tickets that stay valid only while untouched; keeping all coupling in one place means it can be removed in one place; and when those teams ship their own interfaces, our adapters are simply deleted.

Exception (decision 11): Developer 1 may add role checks to two unprotected routes in the compliance module. The change is tiny, it closes a security gap, and it must be written down in the exceptions record in the repository with the reason and date. No other exception is allowed without the same written record.

2.6 Principles that apply everywhere

- **Never lose history.** Changes are added as new records, never written over. Decisions are never edited; a correction or reopening is a new decision that points at the one it replaces.
- **Nothing is permanent except the history.** Flags can be reassessed; rejected leads can be re-reviewed; cleared checks can be reopened.
- **Manual first, automation later.** Every qualification result and check records who or what decided, on what evidence, and whether a person or a computer did it.
- **A problem stays where it happened.** Deal and buyer problems attach to the deal or buyer, never to the customer.
- **The CRM records outcomes; other teams own their decisions.**
- **The server is the authority.** Allowed moves, roles, masking and validation are enforced on the server. The frontend asks the server what is allowed instead of keeping its own copy.
- **Honest placeholders.** Anything not yet real (a stub provider, a pass-through scanner) is clearly labelled as such on screen and in data.

3. Target architecture

3.1 One company, one journey, three gauges, many deals

MAIN JOURNEY · one per company, moves forward only

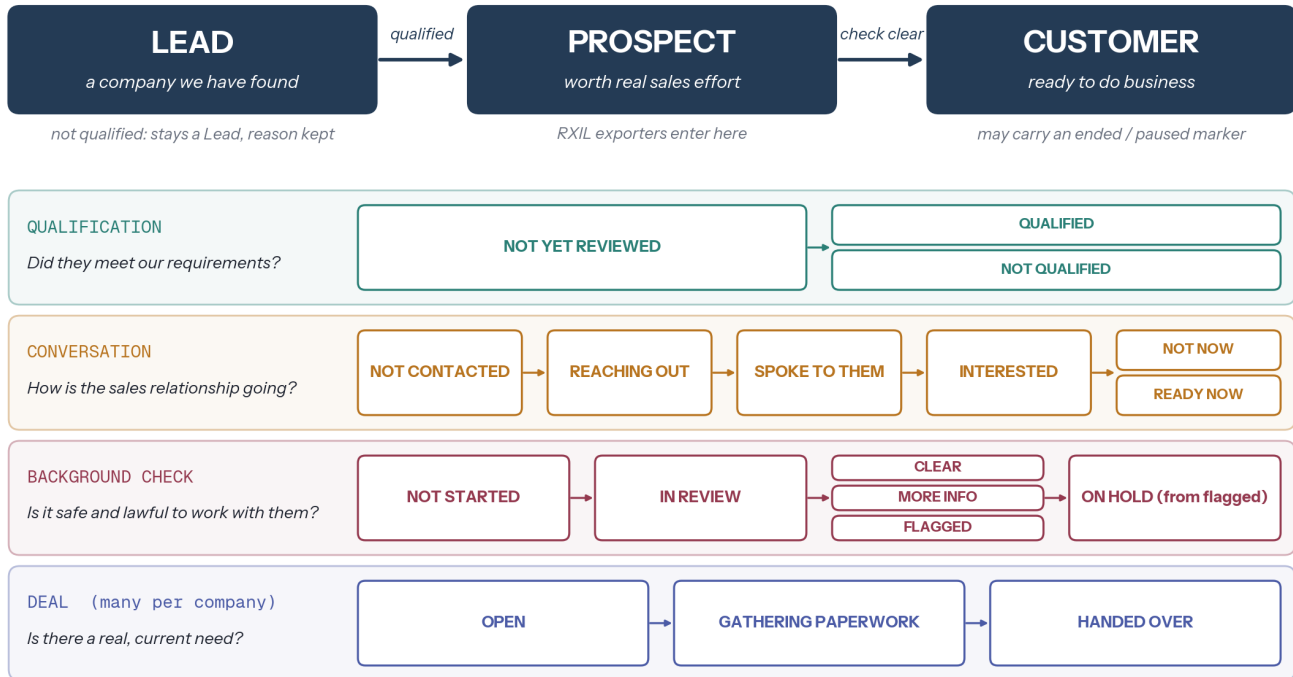


Figure 1. The whole model at a glance. Each row changes independently of the others.

Every company has exactly one record. On it sit the main journey and the current value of each gauge. Deals hang off the company. Every change to any of these writes a row to one shared history log, so the full story of a company can always be shown.

3.2 The main journey

Stage	Meaning	How a company gets here	Moved by
LEAD	A company we have found. No serious sales intent yet.	Bulk import, manual entry, or any other source. Every new company starts here unless it comes from RXIL.	Created by OPERATIONS, COMPLIANCE or ADMIN staff
PROSPECT	Worth real sales effort. Contacting starts here.	Qualification outcome recorded as QUALIFIED; or arrival from RXIL (decision 7).	Automatic when qualification is recorded as QUALIFIED
CUSTOMER	Ready to do business with us.	The company is a Prospect and its background check is CLEAR (decision 2). Whichever happens second triggers the move.	Automatic; announces "became customer" to other teams

Rules. The journey only moves forward. A company that fails qualification stays a LEAD with its reason kept and is never deleted. Leaving or pausing a relationship does not move the journey backwards; instead a separate **marker** is set (decision 3):

Marker	Meaning	Rules
none	Normal (default).	
PAUSED	Relationship temporarily on hold for business reasons (not compliance).	Reason required. Set and cleared by OPERATIONS, COMPLIANCE or ADMIN. Recorded in history.
ENDED	Relationship over.	Reason required. Can be cleared if the relationship restarts. Recorded in history. Ended companies are hidden from default working lists but remain searchable.

Compliance holds are **not** markers. A compliance concern is recorded on the background-check gauge (FLAGGED / ON HOLD). The marker is only for commercial pauses and endings.

3.3 The gauges and the deal in detail

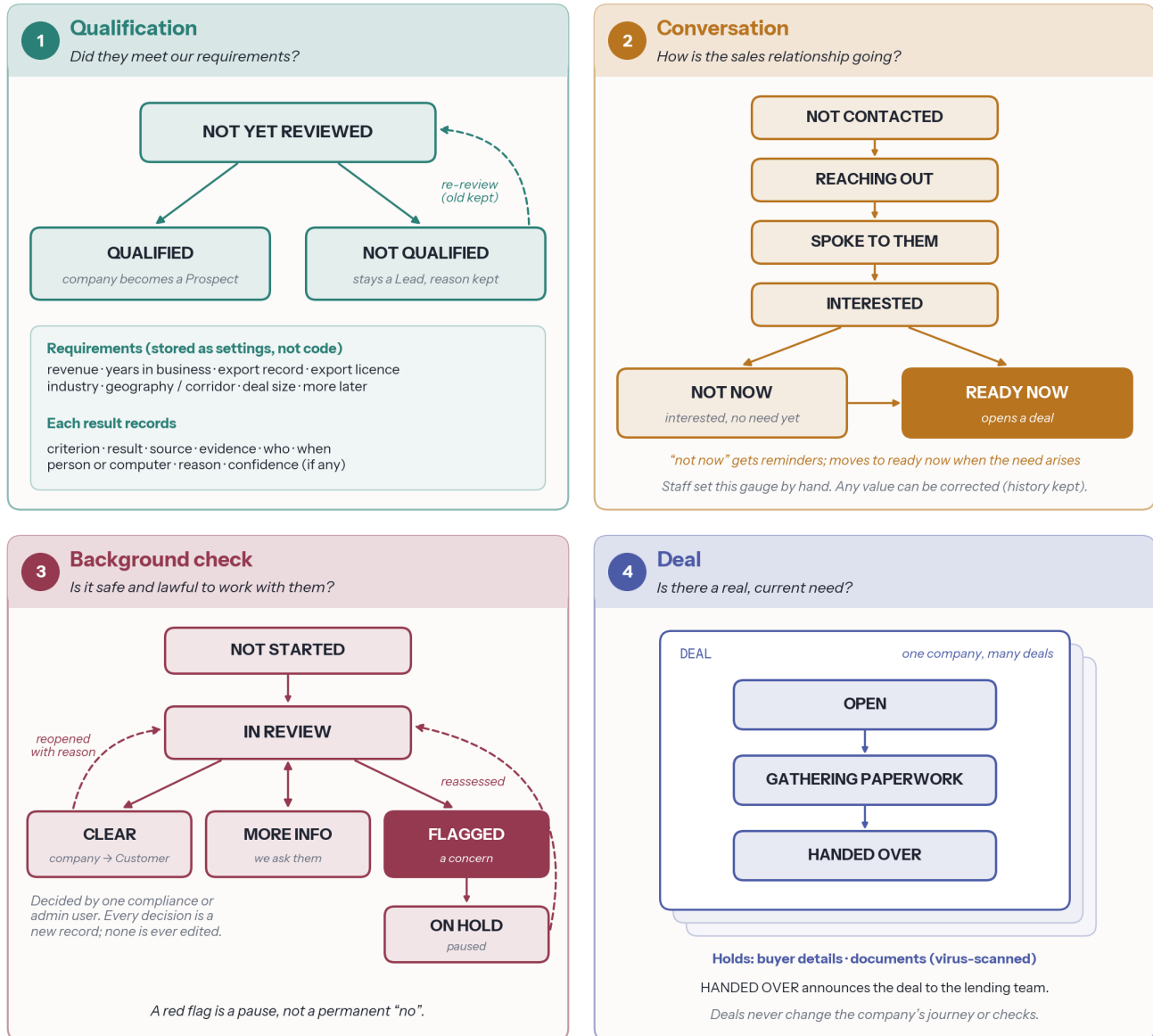


Figure 2. The three gauges and the deal. Dashed arrows are re-reviews and reopenings, which always add a new record.

Gauge 1: Qualification ("Did they meet our requirements?")

Value	Meaning	Next allowed
NOT_YET_REVIEWED	Default for every new lead.	QUALIFIED, NOT_QUALIFIED
QUALIFIED	Meets requirements. The company becomes a Prospect.	(final in the prototype)
NOT_QUALIFIED	Does not meet requirements. Stays a Lead; reason codes and note kept.	QUALIFIED or NOT_QUALIFIED again, through a re-review (decision 8)

- **Criteria are settings, not code.** Each criterion has a key, label, kind (number threshold, yes/no, or list of allowed values), threshold or allowed values, whether it is required, whether it is active, and a version number. ADMIN users manage them. Initial set: revenue, years in business, export history, export licence, industry,

geography / corridor, and deal size. The CEO gave revenue of at least \$100M and at least five years in business as examples; the actual thresholds are set in the settings, not in code.

- **Each result records** criterion and version, result (pass, fail, unknown), source (manual, import, RXIL, automated), evidence, who, when, manual or automated, reason, and confidence if any.
- **The outcome is a person's decision.** After recording results, a reviewer records the overall outcome. The screen suggests QUALIFIED when every required criterion passes, but the person decides. NOT_QUALIFIED requires at least one reason code (for example `revenue_below_threshold`) and may add a note.
- **Re-review** adds new results and a new outcome; old ones stay visible.
- **RXIL companies** arrive with qualification recorded as QUALIFIED, source RXIL (decision 7).
- **Who:** OPERATIONS, COMPLIANCE and ADMIN record results and outcomes; ADMIN manages criteria.

Gauge 2: Conversation ("How is the sales relationship going?")

Value	Meaning
NOT_CONTACTED	No outreach yet. Default for new Prospects.
REACHING_OUT	We have tried to make contact.
SPOKE_TO_THEM	Two-way contact has happened.
INTERESTED	They are interested in working with us.
NOT_NOW	Interested, but no need yet. Kept warm with a check-back follow-up date (required).
READY_NOW	They need financing now. The screen offers to open a deal; opening a deal also sets this.

- Tracked from PROSPECT onward. Set by hand by OPERATIONS, COMPLIANCE or ADMIN; any value can be changed to any other (it is a judgement), and every change is written to history.
- Contacts and the activity log (calls, emails, meetings, notes) already exist and are kept. A follow-up is an activity with a due date; completing it adds a new completion record, because activities are locked. The Follow-ups screen shows every due and overdue follow-up to the whole team.

Gauge 3: Background check ("Is it safe and lawful to work with them?")

From	To	Who	Needs
NOT_STARTED	IN_REVIEW	OPERATIONS, COMPLIANCE, ADMIN; or automatic when RXIL results arrive	-
IN_REVIEW	CLEAR	One COMPLIANCE or ADMIN user	Risk rating; no checks still pending; all eight screening items answered; evidence recorded
IN_REVIEW	MORE_INFO	COMPLIANCE, ADMIN	Note of what is needed
MORE_INFO	IN_REVIEW	OPERATIONS, COMPLIANCE, ADMIN	Note of what arrived
IN_REVIEW	FLAGGED	COMPLIANCE, ADMIN	Reason
FLAGGED	ON_HOLD	COMPLIANCE, ADMIN	Reason
FLAGGED or ON_HOLD	IN_REVIEW	COMPLIANCE, ADMIN (reassessment)	Reason
CLEAR	IN_REVIEW	One COMPLIANCE or ADMIN user (reopen, decision 5)	Reason

- Every move is a new, locked **decision record** (who from the login session, when, reason, the decision it supersedes) and a history row. Nothing is edited. This keeps the database's existing rule that reviews can never be changed, while still letting the gauge move.
- At the moment of each decision, the IDs of the documents and check results it relied on are stored with it (an evidence snapshot), so later uploads cannot silently change what the decision rested on.
- **Risk rating** uses one scale in the CRM: LOW, MEDIUM, HIGH, CRITICAL (decision 6). It is set by compliance as part of the decision. CRITICAL must look different on screen.

- **Inputs:** verification results (manual entry and RXIL for the prototype; Middesk, Trulioo and Sumsbub stay disconnected until credentials and a later phase), plus the existing eight-item screening checklist (for example website reviewed, bank statements reviewed), whose full history is shown.
- A company that becomes FLAGGED stays a Customer if it already is one; the flag is visible and blocks deal handovers (assumption A5).

The deal ("Is there a real, current need?")

Value	Meaning	Rules
OPEN	A current need has been identified.	Opened by OPERATIONS, COMPLIANCE or ADMIN; sets conversation to READY_NOW.
GATHERING_PAPERWORK	Documents and buyer details are being collected.	Documents attach to the deal.
HANDED_OVER	Complete; passed to the lending team.	Only when the company is a CUSTOMER and its check is CLEAR. Announces "deal handed over".
WITHDRAWN	The deal fell through.	Reason required (assumption A7).

A company can have any number of deals over time. Each deal carries its buyer's details (name, country, identifiers). Checks about a buyer are recorded against the buyer, never against the company (decision 9). Transactions and how they perform belong to the lending side.

3.4 Paperwork: documents and storage

Documents belong either to a company (company-wide paperwork) or to a deal (paperwork for that deal). Each document records its category, type, source, file name, file type, size, who uploaded it and when, and a scan status. Only a **relative storage key** is stored, never a cloud address, so the storage provider can change without touching records. Key shape: {env}/{owner_type}/{owner_id}/{source}/{document_id}{ext}; the original file name and type are kept as details, never in the key.

Category	Belongs to	Typical contents
ENTITY_KYC	Company	Incorporation certificate, PAN, GSTIN, IEC, board resolution, owner declaration, director ID
COMPLIANCE_SCREENING	Company	Sanctions, politically-exposed-person and adverse-media evidence
COMPANY_MARKET_REVIEW	Company	Generated analysis (future; source SYSTEM)
BANKING	Both	Company bank statements; per-deal FIRC and e-BRC
INSURANCE	Both	Annual policy; per-shipment marine certificate
PRE_SHIPMENT	Deal	Proforma invoice, purchase order, sales contract, letter of credit
SHIPPING	Deal	Bill of lading or electronic bill of lading, packing list, vessel report
CUSTOMS_AND_REGULATORY	Deal	Shipping bill, customs declaration, GST returns, tax residency
BUYER	Deal	Buyer rating report, buyer KYC
OTHER	Both	Anything else; stops people filing in the wrong category

- Categories are a fixed list checked by the server. Document types within a category are settings (seeded configuration), so new types need no code change. Screens show only the categories that belong where the user is (company page or deal page).
- Source values: RXIL, EXPORTER_UPLOAD (sent by the exporter, uploaded by staff), INTERNAL, SYSTEM.
- **Scan step:** every upload lands as PENDING_SCAN and cannot be opened. A clean result makes it AVAILABLE; an infected or failed scan makes it QUARANTINED or SCAN_FAILED, never served, and raises an alert. In the prototype the scanner is a clearly labelled pass-through. A real scanner is required before real documents (section 7.6).
- **Storage:** local disk behind a storage interface for the prototype and for developers and tests. AWS S3 (with Object Lock for seven-year retention and KMS encryption) is added later as a second implementation of the same interface; local disk stays for development.

3.5 Customer, deal and buyer are separate

If this goes wrong...	It is recorded on...	The customer's record...
A buyer is risky or fails a check	The buyer (a verification result about the buyer)	is unchanged
A deal falls through	The deal (WITHDRAWN with reason)	is unchanged
A transaction performs badly	The lending / ledger side (outside the CRM)	is unchanged
The company itself raises a concern	The company's background check (FLAGGED / ON HOLD)	shows the flag; new deal handovers are blocked

3.6 What the CRM hands over, and to whom

Team or system	Owns	What the CRM gives them	How
Compliance team	Running checks, sanctions logic	Company details; the CRM records their outcomes	Verification adapters (manual, RXIL now; real providers later)
Lending (underwriting)	Whether to fund a deal; transactions	The deal, its buyer and its documents	Event: deal.handed_over

Team or system	Owns	What the CRM gives them	How
Customers team	The customers module used by payments and compliance	Notice that a company became a customer	Event: company.became_customer (they build the receiver later, decision 10)
Ledger	Money records	Nothing directly in the prototype	-
RXIL (partner)	Its own filtering and checks	Nothing; RXIL sends to us	Intake of exporters and results (format provisional)

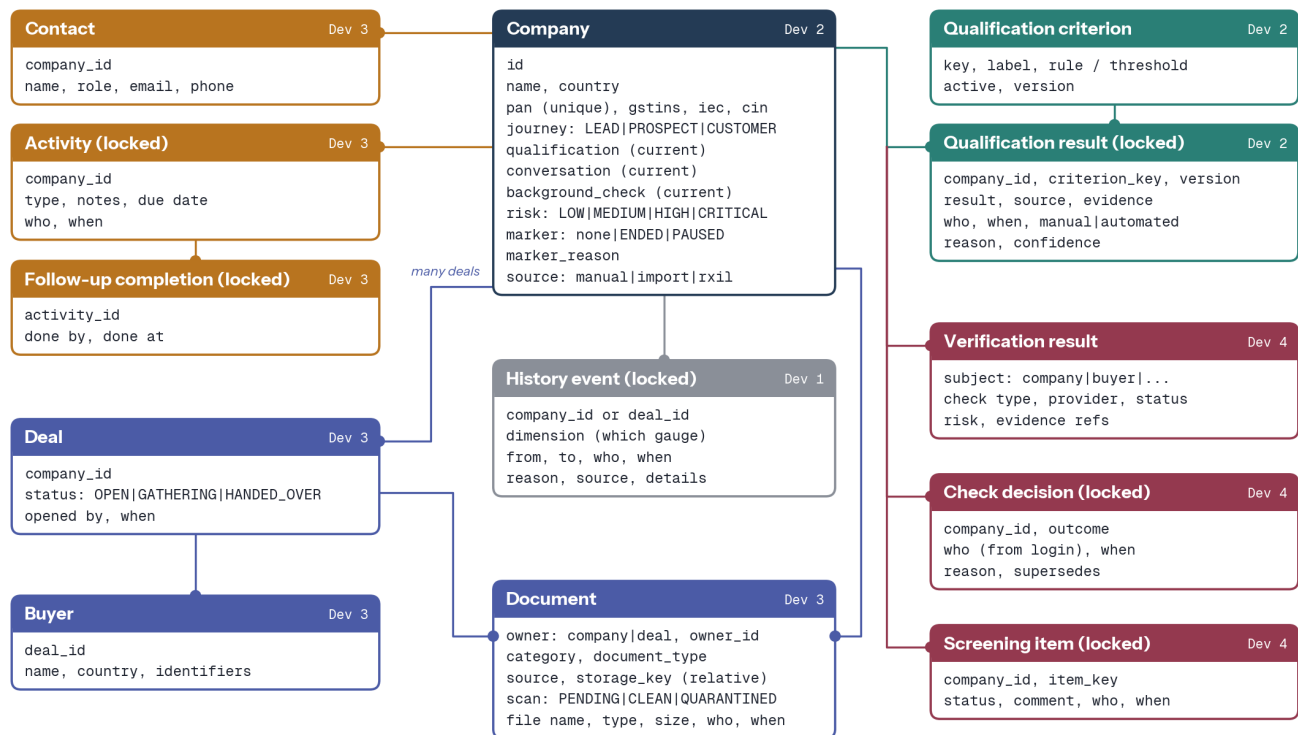
Events. Each event is announced on the existing in-memory event bus after the matching history row is written. The history row is the source of truth; the announcement is best effort. Envelope: event name, event ID, time, company ID, deal ID (if any), who caused it, and a small payload. Payloads: company.became_customer carries company ID, name, country, PAN, GSTINs, risk rating and the clearing decision ID; deal.handed_over carries deal ID, company ID, buyer details and the list of document IDs. Developer 4 publishes the exact format in Week 0.

3.7 Roles and what each can do

Action	OPERATIONS	COMPLIANCE	ADMIN	DEVELOPER	API_USER
See companies, contacts, deals, history	Yes	Yes	Yes	Read only	No
See full PAN / GSTIN	Masked	Yes	Yes	Masked	No
Search by full PAN	Yes	Yes	Yes	No	No
Create and edit companies, contacts, activities	Yes	Yes	Yes	No	No
Record qualification results and outcomes	Yes	Yes	Yes	No	No
Manage qualification criteria	No	No	Yes	No	No
Set conversation, open and move deals, upload documents	Yes	Yes	Yes	No	No
Set ended / paused marker	Yes	Yes	Yes	No	No
Start a background check; answer MORE_INFO	Yes	Yes	Yes	No	No
Background-check decisions, reopen, risk rating, screening items	No	Yes	Yes	No	No
Read audit trails and workflow screens	No	Yes	Yes	No	No

Roles are set by the first-admin command (task L1-09). A new public sign-up receives API_USER, which can reach nothing in the CRM once Developer 1's fixes land. Who "owns" a company (a relationship manager) is not decided for the prototype; everyone with CRM access sees all companies (decision 12).

3.8 Data model



Dot = the "many" end of a link. All links are real database links from day one. "Locked" = rows can be added, never changed or deleted. Field names are a proposal; Week 0 contracts fix the final names. Verification result already exists; the others are new or reshaped.

Figure 3. Proposed records and links. Colours match the owning developer's area.

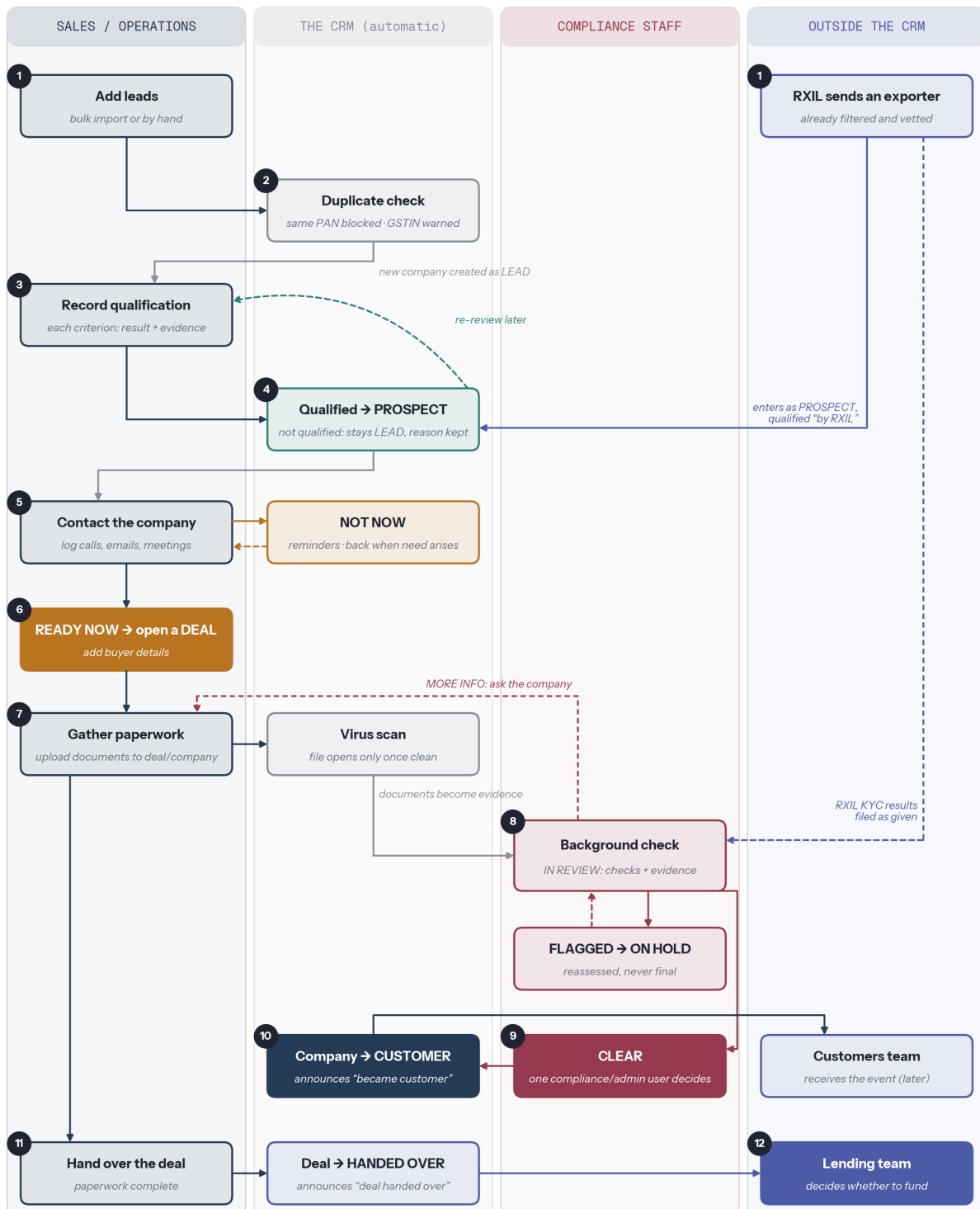
- The company record holds the **current** value of each gauge for fast lists and filters; the history log holds every change. Developers must update both in the same transaction.
- Every child record has a real database link to its parent from day one; records pointing at companies that do not exist become impossible.
- PAN is unique across companies (a second company with the same PAN cannot be saved). A company can hold several GSTINs (one per state); a GSTIN already held by another company produces a warning, not a block (decision 4). A GSTIN contains the company's PAN in characters 3 to 12, so the two are cross-checked.
- The existing verification-results table, the manual and RXIL adapters, the screening-checklist table and the shared "cannot be changed" database protection are kept and reused.

3.9 Three example companies

	Company A	Company B	Company C
Journey	PROSPECT	CUSTOMER	PROSPECT
Qualification	QUALIFIED	QUALIFIED	QUALIFIED
Conversation	NOT_NOW	READY_NOW	READY_NOW
Background check	NOT_STARTED	CLEAR	FLAGGED
Deals	none	two, one handed over	one open, cannot be handed over

All three are normal. None could be described by a single status, which is the reason for the design.

4. The CRM flow, end to end



*Throughout: every change is written to the history log · compliance can reopen a decision with a reason
an ended / paused marker (with reason) can be set at any time · a Customer can open new deals at any time*

Figure 4. Who does what, from finding a company to handing over a deal.

4.1 The main path, step by step

Step	What happens
1 Add leads	Staff import a list or add companies by hand. RXIL exporters arrive through the RXIL intake.
2 Duplicate check	The system blocks a PAN that already exists and warns on a GSTIN already held elsewhere. Bulk import produces a report of accepted, rejected and possible-duplicate rows. New companies start as LEAD, qualification NOT_YET_REVIEWED.
3 Qualification	Staff research the company and record a result for each criterion with evidence. Then they record the outcome.
4 Outcome	QUALIFIED moves the company to PROSPECT. NOT_QUALIFIED keeps it a LEAD with reason codes. RXIL companies arrive directly as PROSPECT, qualified by RXIL.
5 Conversation	Staff contact the company and log activity, moving the conversation gauge. NOT_NOW needs a check-back date and shows on the Follow-ups screen until then.
6 Deal	When the company is READY_NOW, staff open a deal and add the buyer's details.
7 Paperwork	Staff upload documents to the deal or the company. Each file is scanned before it can be opened.
8 Background check	Staff start the check (usually once a deal opens); RXIL results start it automatically. Compliance reviews checks, screening items and documents. MORE_INFO sends a request back to sales; FLAGGED and ON_HOLD are reassessed later.
9 Clear	One compliance or admin user records CLEAR with a risk rating, reason and evidence snapshot.
10 Customer	The company becomes a CUSTOMER automatically and "became customer" is announced.
11 Hand over	When paperwork is complete, staff hand over the deal (only allowed for a Customer whose check is CLEAR).
12 Lending	"Deal handed over" is announced to the lending team, who decide whether to fund.

4.2 Other paths

Situation	What happens
Lead fails qualification	Stays LEAD, NOT_QUALIFIED, reasons kept. Can be re-reviewed later with new results; the old ones stay in history.
"Interested, but not now"	Conversation NOT_NOW with a check-back date. Returns to READY_NOW when the need arises.
Compliance needs more	Background check MORE_INFO; sales collects it; the check returns to IN_REVIEW.
Red flag	FLAGGED, possibly ON_HOLD. Reassessed back to IN_REVIEW and possibly CLEAR later. Deal handovers are blocked meanwhile.
New information about a cleared company	One compliance or admin user reopens (CLEAR → IN_REVIEW) with a reason. The company stays a Customer.
Existing customer has a new need	Staff open another deal. The company record and gauges are untouched.
Deal falls through	Deal WITHDRAWN with a reason. The company is untouched.
Relationship paused or ended	Marker PAUSED or ENDED with a reason. Journey unchanged; reversible.

5. Where we stand today

This section summarises a read-only audit of the codebase carried out on 25 September 2026. Evidence keys such as [E11] point to Appendix A, which lists the exact files and lines.

5.1 Build and tests

Check	Result
Backend test suite	2,797 passed, 22 failed, 5 errors, 6 skipped (13m 41s). All 27 red items are payments / FX tests that get 404 because those routes are not switched on in this checkout. This is the baseline : any other failure is new. [E1]
Frontend build	Passes. One warning: main bundle 517 kB (guideline 500 kB). Frontend unit tests: 4 files, 24 tests, all pass. [E2]
Generated API types	Git-ignored, so a fresh clone cannot type-check until they are generated with the backend environment. Fixed by task L1-03. [E3]
Database changes (migrations)	71 revisions, one head (onboarding_0012_risk_critical), no breaks. Next free onboarding number: 0013 . Two duplicate numbers exist in other modules (compliance_0002, rails_0003); leave them alone. [E4]

5.2 What exists, what we keep, what we build

Part of the target	Today	Keep	Change or build
Company record and journey	Partial	Profile table, company IDs, lead source, lifecycle history table	Replace 10 statuses with 3 stages; move name and country into the CRM; add real links; make PAN unique; add marker
Qualification	Missing	The "add a row, never overwrite" pattern	Everything: criteria, results, reasons, re-review, bulk import, duplicate matching
Conversation	Partial	Contacts, activity log, follow-up due dates	Gauge values; follow-up completion; reminders screen; check that the company exists
Deal	Missing	-	The whole record, its stages and its buyer
Paperwork	Missing	Legacy document type names as a reference only	Document records, storage, scanning
Background check	Partial	Verification results, manual provider, screening checklist, reviewer taken from login	The gauge; superseding decisions (reopen/reassess); stop result overwrites; honest placeholders
Customer vs deal vs buyer	Partial	Verification results can already be about a buyer	Buyer and deal records
Handovers	Missing	The in-memory event bus exists	Events from the CRM (it publishes none today)
History	Partial	Shared "cannot be changed" protection (22 tables); lifecycle history table	One history log for every gauge and deal, with a read route

5.3 Security status

Problem	Status	What it means today	Fixed by
Approver can record a human approval as "system"	Open	A compliance user can approve an onboarding case with no human recorded; a test asserts this. [E29]	L1-05
Notifications need no login	Open	Anyone on the network can list and trigger notifications. [E27]	L1-06
Routes that check login but not role	Open	Any account, including a fresh sign-up, can read audit trails and use six other routes. [E26]	L1-07, L1-08
No first administrator	Open	Roles can only be granted by editing the database. [E25]	L1-09

Problem	Status	What it means today	Fixed by
Full-PAN search by DEVELOPER	Open	A developer account can learn which company holds a PAN. [E30]	L1-10
Sign-up with a chosen role	Fixed	Sign-up always gives the lowest role. [E25]	-
Provider field could load code	Fixed	Only "manual" and "rxil" accepted. [E28]	-
Tax IDs masked only in the browser	Fixed	Masked on the server now. [E30]	-

5.4 Things that look finished but are not

- "Run available checks" creates placeholder rows that stay pending forever. [E18]
- A manual "sanctions: passed" can be recorded with no evidence and looks like any other result. [E18]
- The bank-activity panel always shows 0 connected accounts; nothing ever writes findings. [E19]
- The only sanctions screening is an exact-name match on payments, not on companies. [E20]
- Dashboard is a placeholder; Follow-ups is marked "soon". [E39]
- A company can be walked from LEAD to ONBOARDED with an empty checklist and no checks. [E38]
- Provider polling overwrites a result's status and risk even after it has been reviewed. [E17]

5.5 Collisions to know about

- **Names:** two classes called ComplianceCase (cases and compliance modules), two Customer classes, two CaseRepository classes, and an onboarding Case with its own 12-state machine. Always use the full module path when importing. [E23]
- **"Checklist":** the existing "screening checklist" is a compliance list and belongs to the background check (Developer 4). The sales qualification gauge is always called **qualification**. [E15]
- **Customers module:** tables only, an empty route list, and no code that creates a customer. 0 of 540 CRM company IDs match it. We do not write to it (decision 10). [E12][E22]
- **Case engine:** built but not switched on; nothing outside its own module uses it. Not used in the prototype (assumption A6). [E21]
- **Risk scales:** four different ones exist; the CRM uses LOW / MEDIUM / HIGH / CRITICAL only. [E40]
- **Module rule loophole:** the import checker only protects onboarding's internals, and the KYB module names onboarding's adapter folder in its own allow-list. Do not extend either. [E36]

5.6 What the old ten statuses mean in the new model

We start fresh (decision 1), so no data is converted. This table is for reading the existing code and tests while replacing them.

Old status	New home
LEAD	Journey LEAD; qualification NOT_YET_REVIEWED
CONTACTED	Conversation REACHING_OUT or SPOKE_TO_THEM
DATA_COLLECTION	Journey PROSPECT; deal GATHERING_PAPERWORK (or background check MORE_INFO when sent back)
VERIFICATION_IN_PROGRESS, COMPLIANCE_REVIEW	Background check IN_REVIEW
ONBOARDED	Background check CLEAR, and therefore journey CUSTOMER
FINANCING_ELIGIBLE	Removed: a lending decision, outside the CRM
ACTIVE	Journey CUSTOMER
SUSPENDED	Background check ON_HOLD, or marker PAUSED for a commercial suspension
OFFBOARDED	Marker ENDED

6. Decisions

6.1 The twelve decisions settled for the prototype

#	Question	Decision	Why	Effect and owner
1	Existing data	Start fresh with sample data.	Local data is mostly test leftovers; converting it was the hardest part of Developer 2's work.	No data migration or clean-up. Real links can be added from day one. Developer 2 writes a sample-data script.
2	When a company becomes a Customer	When its background check is CLEAR.	Matches the CEO's wording (checks completed, ready to do business); the CRM can see it directly.	Developer 4 signals CLEAR; Developer 2 moves the company (whichever of Prospect/CLEAR comes second).
3	Ended or paused relationships	A marker (ENDED / PAUSED) with a reason, separate from the journey.	Keeps three clean stages; reversible; records why.	Small task for Developer 2.
4	Duplicate tax IDs	Block a duplicate PAN; warn on a duplicate GSTIN.	A company has one PAN but may hold several GSTINs, one per state.	Database rule on PAN; warnings in forms and bulk import (Developer 2).
5	Reopening a cleared check	One COMPLIANCE or ADMIN user, with a required reason.	The reason and the permanent record are the safeguard; two approvers would stall a prototype.	Superseding decision records (Developer 4).
6	Official risk scale	LOW / MEDIUM / HIGH / CRITICAL.	The onboarding scale already has CRITICAL. "Prohibited" is an outcome, recorded as FLAGGED.	Developer 4 uses it everywhere in the CRM; other modules' scales untouched.
7	Where RXIL exporters enter	As Prospects, qualification marked "by RXIL".	RXIL has already filtered them; we never recompute RXIL's work.	RXIL company intake (Developer 2); RXIL results into the background check (Developer 4).
8	Re-qualifying a rejected lead	Allowed; new results added, old ones kept.	Revenue grows and criteria change; history shows why the answer changed.	Developer 2.
9	Transactions and buyers	The CRM keeps buyers on deals; lending owns transactions.	The CRM gathers buyer paperwork anyway; the CEO placed transactions outside the CRM.	Buyer on deal (Developer 3); buyer checks on the buyer (Developer 4).
10	Telling other teams about a new customer	Announce an event only, for now.	The customers module has no way to create a customer and we may not edit it.	Event format by Developer 4; the customers team builds the receiver later.
11	Unprotected compliance routes	Fix them as a documented one-off exception to the module rule.	A tiny change; an open route is worse than bending the rule once, on the record.	Developer 1.
12	Who sees full tax IDs	COMPLIANCE and ADMIN; sales staff see them masked.	Company ownership (relationship managers) can wait until after the prototype.	Developer 1.

6.2 Earlier decisions: what still holds and what changed

The previous plan (v1.1) recorded decisions D1 to D17 and a work register (A, B and C items). Their fate:

Earlier item	Status now
D1 Role matrix (OPERATIONS and up write; COMPLIANCE/ADMIN review; never DEVELOPER/API_USER writes)	Holds; see 3.7
D2 Follow-ups visible to the whole team	Holds
D3 First ADMIN via an idempotent seed command	Holds; task L1-09
D4 Providers stored lowercase, displayed uppercase	Holds

Earlier item	Status now
D5 Approvals through the case engine, single approver	Single approver holds (decision 5). Case engine deferred (A6)
D6 Announce through the existing event bus	Holds; section 3.6
D7 Stage history in onboarding_event	Replaced: one generalised history log (task L1-11)
D8 Storage: local disk first, then AWS S3; relative keys only	Holds; S3 after the prototype
D9 Scan step now, real scanner later	Holds; real scanner gates real documents
D10 Reopen by one user with a reason, as a new record	Holds (decision 5)
D11 Temporal / legacy onboarding descope	Holds; never build on it
D12 Funding decided per invoice	Moved out of the CRM: lending owns funding; the CRM has deals
D13 Record where each item came from	Holds for companies, results, deals and documents
D14, D15 Ten document categories, types as settings	Holds; "invoice" is now "deal"
D16 Generated documents with source SYSTEM	Holds for later; not built in the prototype
D17 Evidence snapshot at decision time	Holds; built into background-check decisions
Status FINANCING_ELIGIBLE	Removed
C8 to C13 invoice layer (invoice, funding case, invoice checks)	Removed from the CRM. Buyer kept on deal; RXIL intake kept
A1.x security items	A1.1, A1.3, A1.5 done; the rest are Developer 1's tasks
A2.1, A2.2 history gaps	Done: both tables now use the shared protection [E13]
A3.1 broken frontend build	Done: build passes [E2]

6.3 Planning assumptions

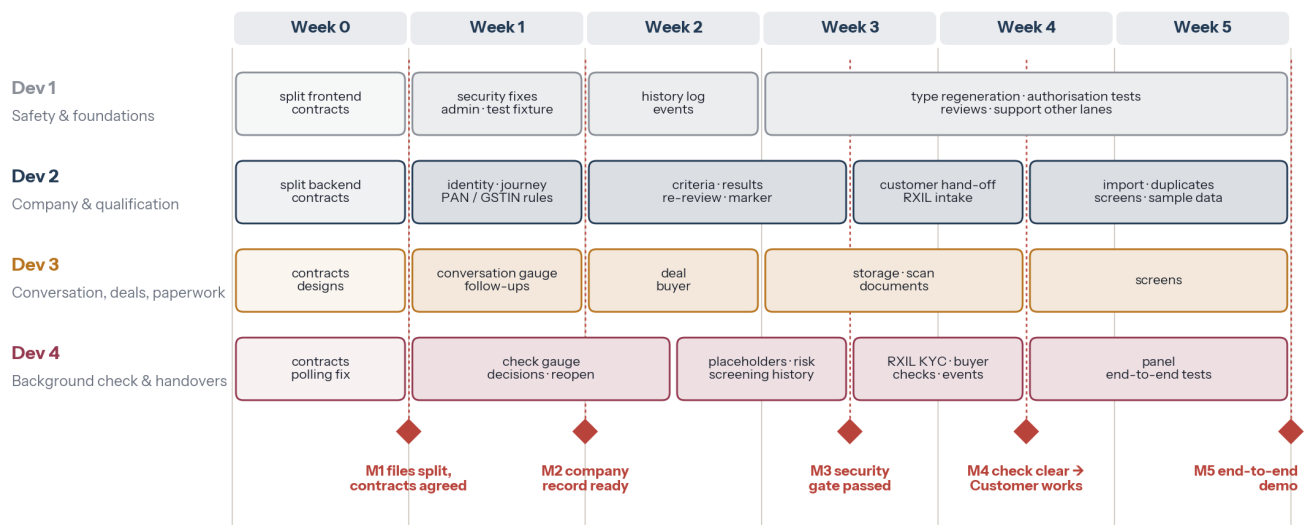
These fill gaps the decisions did not cover, so developers never have to stop and ask. Each is a recommended default. To change one, edit it here and tell the owner.

#	Assumption	Owner
A1	A company moves to CUSTOMER when it is a Prospect and its check is CLEAR, whichever happens second.	Dev 2, Dev 4
A2	The journey only moves forward. Once QUALIFIED, qualification is not re-opened in the prototype; re-review applies to NOT_QUALIFIED leads.	Dev 2
A3	CLEAR requires a risk rating, no checks still pending, all eight screening items answered, and an evidence snapshot.	Dev 4
A4	The conversation gauge applies from PROSPECT onward; staff may change it to any value; NOT_NOW needs a check-back date.	Dev 3
A5	A deal can be handed over only when the company is a CUSTOMER and its check is CLEAR. A flagged customer stays a customer.	Dev 3, Dev 4
A6	Background-check decisions are recorded in the CRM. Connecting the separate case engine is left until after the prototype; event formats allow it to subscribe later.	Dev 4
A7	A deal can be WITHDRAWN with a reason.	Dev 3
A8	A background check is started by staff (usually once a deal opens) or automatically when RXIL results arrive.	Dev 4
A9	The prototype uses local-disk storage and a clearly labelled pass-through scanner.	Dev 3

#	Assumption	Owner
A10	Public sign-up stays, giving the lowest role, which reaches nothing in the CRM; a setting can switch sign-up off.	Dev 1
A11	ENDED companies are hidden from default working lists but stay searchable.	Dev 2
A12	The first-admin command can also promote an existing user to a role.	Dev 1
A13	Middesk, Trulioo and Sumsbub stay disconnected in the prototype; manual entry and RXIL are the providers.	Dev 4
A14	Bulk import accepts CSV with a fixed column template; a report lists accepted, rejected and possible-duplicate rows.	Dev 2

7. Implementation plan

7.1 Timeline



Indicative only. Week 0 is about three working days. Bars show the order of work within each lane; milestones are the points where lanes must meet.

Figure 5. Indicative timeline for four developers. Sizes: S = under a day, M = a few days, L = a week or more.

The critical path runs through Developer 2: every gauge, deal and screen hangs off the company record. Developer 1 has spare capacity from Week 3; if Developer 2 falls behind, Developer 1 takes the pipeline and list screens (L2-14) or the import screen, with Developer 2 reviewing.

7.2 Week 0 (about three days)

Four files today carry most of every developer's changes: the company routes, the company request and response shapes, the profile service, and the company detail page. If they are not split first, "working in parallel" becomes "waiting in a queue". Week 0 splits them **without changing behaviour** and fixes the contracts everyone builds against.

Who	Week 0 work
Dev 2	Split backend: contacts and activities routes and shapes into their own files (to Dev 3); screening, verification and bank-activity routes and shapes into their own files (to Dev 4); lift the compliance gate out of the profile service into a Dev 4 service; split the constants file into journey values and screening items.
Dev 1	Split frontend: the company detail page becomes a shell with four panels (Company: Dev 2; Conversation and Deals: Dev 3; Background check: Dev 4); split API and hooks files per developer; take ownership of generated types. Open the migration register.
Dev 3	Write the engagement, deal and document/storage contracts; sketch screens.
Dev 4	Write the background-check and event contracts; fix provider polling overwriting reviewed results (own file, no conflict).

Contract	Owner	Contents	Used by
History row	Dev 1	company ID or deal ID, dimension (which gauge), from, to, who, when, reason, source, details	All
Event envelope	Dev 1	name, ID, time, company ID, deal ID, actor, payload	All
Migration register	Dev 1	Numbers and merge order (7.4)	All
Company record	Dev 2	ID, name, country, PAN, GSTINs, IEC, CIN, journey values, marker, current gauge fields	All

Contract	Owner	Contents	Used by
Criterion result	Dev 2	Shape in section 3.3	Dev 4 (RXIL), later automation
Engagement values	Dev 3	Values and rules in section 3.3	Dev 2 (lists), Dev 4
Deal and buyer	Dev 3	Fields and stages in 3.3 and Figure 3	Dev 2, Dev 4
Storage interface	Dev 3	upload, get download link, delete; relative keys; scan status	Dev 4 (evidence)
Background-check values and moves	Dev 4	Table in 3.3, including who and what each move needs	Dev 2 (Customer move), Dev 3 (handover guard)
Handover events	Dev 4	company.became_customer and deal.handed_over payloads (3.6)	Other teams

Week 0 is finished when the backend suite is still at the baseline (2,797 passing, only the 27 known payments/FX failures), the frontend builds, every contract above is merged as a short document or typed definition, and every shared file has a named owner (section 8).

7.3 Milestones

Milestone	When	How we know
M1 Files split, contracts agreed	End of Week 0	Section 7.2 exit check passes.
M2 Company record ready	End of Week 2	A company can be created with its own name, moved LEAD → PROSPECT, and duplicate PAN is refused; history rows are written.
M3 Security gate passed	Mid Week 3	Every item in 5.3 marked Open is fixed, with route tests; placeholders in 5.4 are fixed or labelled.
M4 Check clear → Customer works	Mid Week 4	A Prospect whose check is recorded CLEAR becomes a CUSTOMER and "became customer" is announced.
M5 End-to-end demo	End of Week 5	The twelve steps of section 4.1 and the paths in 4.2 run on sample data, covered by an automated end-to-end test.

7.4 Migration register

No.	Owner	Change	Depends on
0013	Dev 1	Generalise the history table: dimension column, optional deal ID	-
0014	Dev 2	Fresh company record: name, country, identifiers, journey (3 values), marker, current gauge fields, PAN unique; real links from contacts, activities, screening items	0013
0015	Dev 4	Background-check decisions (locked, superseding); allow superseding reviews on verification results; risk scale	0014
0016	Dev 3	Follow-up completion (locked)	0014
0017	Dev 2	Qualification criteria and results	0014
0018	Dev 3	Deal and buyer	0014
0019	Dev 3	Documents	0018

- Migrations form one chain. If the merge order changes, the later migration updates its parent pointer (a one-line change) and Developer 1 updates the register.
- Revision IDs must be 32 characters or fewer. Never add values to a database list type inside an autocommit block (this has broken the repository before). Keep exactly one head.
- Because we start fresh, 0014 may drop and recreate the CRM's own tables. It must not touch the legacy onboarding tables or any other module's tables.

7.5 Working rules

- **Tests stay at the baseline.** Only the 27 known payments/FX failures are allowed. Anything else blocks a merge.
- **Every state change writes a history row** in the same transaction, with a reason wherever section 3 requires one, and the actor taken from the login session, never from the request body.
- **Every new route declares its roles** and has a test in the route-authorisation file, including the refusal (403) response in the API description.
- **The server serves allowed moves.** The frontend fetches the allowed journey, gauge and deal moves for the current user instead of keeping hand-copied tables.
- **One owner regenerates API types** (Developer 1), at each milestone and on request.
- **Shared files** change only through a pull request reviewed by the owner (section 8).
- **Contracts change only** with the agreement of the owner and every user of that contract.
- **The module rule holds**, with the single documented exception in 2.5.
- **Placeholders are labelled** on screen and in the data (for example provider "manual", scanner "pass-through").

7.6 Gates

Before...	Must be true	Why
Anyone outside the team uses the prototype	All Developer 1 security fixes; Developer 4 placeholder clean-up (evidence required, honest pending checks, honest bank panel)	Otherwise people can reach what they should not, and a fake "passed" check looks real.
Any real exporter data or real documents	A real virus scanner behind the scan step (AWS GuardDuty Malware Protection for S3 or self-hosted ClamAV); S3 storage with Object Lock and KMS encryption verified	A pass-through scanner is an open upload point; local disk has no retention lock and seven-year AML retention is required.
Production	Field-level encryption / KMS for tax IDs (a platform gap today); real provider credentials	Several tickets assume encrypted storage that does not exist yet.

7.7 Definition of done for every task

- Behaviour matches section 3 and the relevant contract; roles enforced on the server.
- History rows written for every change; locked tables stay locked.
- Unit tests for the rules, route tests for roles, and at least one test for each refusal.
- Screen updated where the task has a user-facing part, with loading, empty and error states.
- API types regenerated by Developer 1 when routes changed.
- Suite at baseline; frontend builds; no new lint or import-rule violations.

8. Coordination

8.1 Shared files and their owners

File (plain name)	Touched by	Owner	Rule
Company routes file	All four	Dev 2	After Week 0, others add their own route files beside it
Company request/response shapes	Dev 2, 3, 4	Dev 2	Split per developer in Week 0
Company profile service	Dev 2, 3, 4	Dev 2	Compliance gate moved to Dev 4 in Week 0
CRM status lists file	Dev 2, 3	Dev 2	Each gauge's values in its owner's file
Onboarding errors, entity index, repository index	All four	Dev 1	Additions only; owner reviews
Onboarding migrations folder	All four	Dev 1	Numbers from the register (7.4)
Route-authorisation test file	All four	Dev 1	Every new route gets a row
Frontend company detail page	Dev 2, 3, 4	Dev 2 (shell)	One panel per developer after Week 0
Frontend API, hooks, types, generated API types	Dev 2, 3, 4	Dev 1	Per-developer files after Week 0
Frontend constants file	Dev 2, 4	Dev 2	Split in Week 0
Sidebar and routes	Dev 1, 3	Dev 1	Dev 3 adds Follow-ups, Deals and Documents entries

8.2 Hand-off points between developers

Between	What passes	Where it is agreed
Dev 4 → Dev 2	"Check became CLEAR" signal; Dev 2 moves the company to CUSTOMER and announces it	Background-check contract; A1
Dev 3 → Dev 4	Document and check-result IDs used as evidence at decision time	Storage interface
Dev 3 ↔ Dev 4	Deal handover guard reads the company's check value; buyer checks attach to the deal's buyer	A5; deal contract
Dev 2 ↔ Dev 4	RXIL package: company part to Dev 2, results part to Dev 4	RXIL format (provisional), one shared parser
Dev 1 → all	History writer, event helper, role checks, generated types	History and event contracts
Dev 2 → all	Company record and IDs everything links to	Company contract

A short daily check-in covers blockers and contract questions. Any contract change is written down before code changes.

9. Developer responsibilities

Each part lists the mission, owned areas, Week 0 contracts, tasks with size and dependencies, and what to watch out for. File paths are relative to backend/app/ unless marked frontend:.

9.1 Developer 1: Safety and shared foundations

Mission. Make the prototype safe to show to anyone, and give the other three the shared tools they build on: the history log, event announcements, role checks, tests, migration order and frontend plumbing.

Owns

Login and roles (api/rest/auth/*, platform/authorization/services.py, platform/authentication/models.py); modules/notifications/api/router.py; modules/audit/api/router.py; api/rest/workflows.py; onboarding case service and schemas/case.py; history entity and repository; onboarding errors, entity and repository indexes; conf/test.py, alembic.ini, importlinter.ini; the route-authorisation test file; frontend src/lib/api/*, src/platform/auth/*, src/routes/*, layout/Sidebar.tsx and generated API types.

Publishes in Week 0

History row shape, event envelope, migration register.

Tasks

ID	Task	Size	Needs	Done when
L1-01	Split frontend shared files: detail page shell with four panels; per-developer API and hooks files	M	Week 0	Build passes, no behaviour change, each panel file has one owner
L1-02	Publish history and event contracts; open the migration register	S	Week 0	Merged and acknowledged by all
L1-03	Make a fresh clone build: generate API types in setup or commit them [E3]	S	-	New clone type-checks with one documented command
L1-04	Test fixture that creates privileged accounts without the sign-up route	S	-	Tests get COMPLIANCE/ADMIN tokens directly
L1-05	Forged approver: human decisions always record the logged-in user; remove caller-chosen "system" source; fix the test that asserts the old behaviour [E29]	S	L1-04	Approving as COMPLIANCE records that user; test asserts it
L1-06	Notifications require login and a staff role [E27]	S	L1-04	No-token requests get 401
L1-07	Role checks on audit and workflow routes [E26]	S	L1-04	API_USER gets 403 on all of them
L1-08	Role checks on the two compliance-module routes, recorded as the one exception (decision 11)	S	L1-04	Exception written in the repository; routes return 403 for API_USER
L1-09	First-admin command: idempotent, reads environment variables, creates first ADMIN and one COMPLIANCE user, can promote an existing user (A12)	S	-	Running twice changes nothing; roles granted without database edits
L1-10	Tax-ID visibility: full for COMPLIANCE/ADMIN, masked for others; DEVELOPER cannot search by full PAN [E30]	S	-	Route tests for each role
L1-11	Shared history log: generalise the history table (0013), writer helper, read route for a company or deal	M	L1-02	Every gauge and deal change can be listed in order with who, when, reason
L1-12	Event helper: announce on the existing bus after the history row, best effort	S	L1-11	Events visible in tests; failures logged, never block the change
L1-13	Sign-up setting (A10): confirm API_USER reaches nothing in the CRM; add a switch to turn sign-up off	S	L1-07	Route test proves API_USER is refused everywhere in the CRM
L1-14	Regenerate API types at each milestone; keep route-authorisation tests complete	cont.	-	Types match the API at M2, M3, M4, M5

ID	Task	Size	Needs	Done when
L1-15	From Week 3: help the critical path (screens from L2-14) as agreed with Dev 2	flex	-	Agreed hand-over of specific screens

Watch out for

- Do L1-04 before L1-05 to L1-08, or the test suite loses its way to get privileged accounts.
- The onboarding case model with the forged approver belongs to the legacy path; fix it, but do not extend it.
- Never add another exception to the module rule without writing it down in the same place as L1-08.

9.2 Developer 2: The company and qualification (critical path)

Mission. Give every company its own proper record and a clear three-stage journey, and build qualification from nothing, so everyone else has something solid to attach to.

Owns

The lifecycle part of `domain/entities/exporter_enums.py`; `exporter_profile.py`, `application/exporter_profile_service.py`, `exporter_profile_views.py` and the profile repository; `api/exporter_router.py` (company routes); the company part of `api/schemas/exporter.py`; `exceptions.py`; the new qualification code; `frontend/constants.ts` (journey part), `types.ts`, `StageChip`, `LifecycleMoveControl`, `list`, `pipeline` and `add-company` pages, and the `detail-page` shell.

Publishes in Week 0

Company record (identity, journey values, marker, current gauge fields); criterion-result shape.

Tasks

ID	Task	Size	Needs	Done when
L2-01	Split backend shared files (routes, shapes, service, constants) as in 7.2	M	Week 0	Suite at baseline; each file has one owner
L2-02	Publish the company and criterion-result contracts	S	Week 0	Merged and acknowledged
L2-03	Company identity: name and country on the company record; search and lists no longer read the legacy table [E11]	M	L2-02	No CRM query touches the legacy onboarding table
L2-04	Three-stage journey replacing the ten statuses across 18 backend files, 5 screens and 3 frontend tests; forward only; server returns allowed moves [E5 to E9]	L	L2-03	Old statuses gone; allowed moves served by the API
L2-05	Fresh start (migration 0014) with real links; sample-data script covering every example in 3.9	M	0013	Clean database builds; sample data loads in one command
L2-06	Tax-ID rules: PAN format and uniqueness; GSTIN duplicate warning; GSTIN-contains-PAN cross-check	S	L2-05	Duplicate PAN refused; GSTIN warning shown; tests
L2-07	Profile edits keep history; fields can be cleared [E31]	S	L1-11	Each edit leaves a history row
L2-08	ENDED / PAUSED marker with reason; hidden from default lists (A11)	S	L1-11	Set, clear, filter, history
L2-09	Qualification criteria as versioned settings, managed by ADMIN, seeded with the initial set	M	L2-05	Criteria editable without code; old results keep their version
L2-10	Qualification results, overall outcome with reason codes, re-review; QUALIFIED moves to PROSPECT	M	L2-09	Rules in 3.3 enforced; history shows every result
L2-11	Customer move: on Dev 4's CLEAR signal, or on qualification if already CLEAR; announce <code>became_customer</code>	S	L4-05, L1-12	M4 passes
L2-12	RXIL company intake: create or match the company as PROSPECT, qualification source RXIL (provisional format)	M	L2-10	A sample RXIL package creates a qualified Prospect
L2-13	Bulk CSV import with template, validation report and duplicate matching on PAN, GSTIN (via PAN), IEC and CIN (A14)	L	L2-06	1,000-row sample imports with a correct report
L2-14	Screens: company list with filters for journey, gauges and marker; three-column pipeline; add company; company panel with qualification	L	L2-04, L2-10	Usable end to end on sample data

Watch out for

- Company identity is the real refactor: search currently joins the legacy table. Do L2-03 before anything else.
- Do not reuse the screening checklist as qualification; it is a compliance list owned by Developer 4.

- Update the current gauge field on the company and write the history row in the same transaction.
- If you fall behind, hand screens to Developer 1 early rather than late.

9.3 Developer 3: Conversations, deals and paperwork

Mission. Track the sales relationship and every deal from first contact to paperwork gathered, and build safe document storage.

Owns

application/exporter_contact_activity_service.py, contact and activity entities and repositories, activity types, the new engagement, deal, buyer, document and storage code and routes (storage replaces the empty integrations/object_storage scaffold); frontend Conversation and Deals panels, Follow-ups page, deal and document screens.

Publishes in Week 0

Engagement values; deal and buyer; storage interface.

Tasks

ID	Task	Size	Needs	Done when
L3-01	Publish engagement, deal/buyer and storage contracts	S	Week 0	Merged and acknowledged
L3-02	Contacts and activities must point at a real company [E16]	S	L2-05	Ghost-company writes refused
L3-03	Conversation gauge on the company, with history (A4)	M	L2-05, L1-11	Values and rules in 3.3; history listed
L3-04	Follow-up completion as a new locked record (0016); Follow-ups screen for the whole team; NOT_NOW requires a check-back date	M	L3-03	Due and overdue lists correct; completing leaves a record
L3-05	Deal record (0018): many per company; OPEN, GATHERING_PAPERWORK, HANDED_OVER, WITHDRAWN (A7); READY_NOW prompts a deal	M	L2-05	Stages and history work; opening a deal sets READY_NOW
L3-06	Buyer details on the deal	S	L3-05	Buyer saved, shown and linked for checks
L3-07	Storage interface and local-disk implementation; relative keys; download through the API or a short-lived link	L	L3-01	Upload and download work; no cloud address stored
L3-08	Scan step: PENDING_SCAN until cleared; labelled pass-through scanner (A9); quarantine and alert paths	S	L3-07	Unscanned files cannot be opened
L3-09	Document records (0019) owned by company or deal; categories checked by the server; types as settings; source	M	L3-07	Invalid category refused; contextual category lists
L3-10	Hand over a deal: guard (Customer and CLEAR, A5); announce deal.handed_over	S	L4-03, L1-12	Guard tested; event carries buyer and document IDs
L3-11	Screens: Conversation panel, Follow-ups page, deal list and detail, document upload and list	L	L3-03 to L3-10	Usable end to end on sample data

Watch out for

- Activities are locked by the database. Never try to update one; add a completion record instead.
- Keep file names and types out of storage keys.
- The scanner is a placeholder; the screen must say so.

9.4 Developer 4: Background checks and handovers

Mission. Run the background-check gauge properly (never stuck, never rewritten), keep customer, deal and buyer problems apart, and define what the CRM hands to other teams.

Owns

application/verification_service.py, api/schemas/verification.py, the verification and screening routes, screening_review_service.py and model, domain/entities/orchestration_enums.py, the manual and RXIL adapters, the adapter registry (domain/workflow_dependencies.py), the compliance gate lifted from the profile service; frontend VerificationSection.tsx and the Background check panel.

Publishes in Week 0

Background-check values and moves; handover event formats.

Tasks

ID	Task	Size	Needs	Done when
L4-01	Publish background-check and handover-event contracts	S	Week 0	Merged and acknowledged
L4-02	Stop provider polling overwriting a reviewed result's status and risk [E17]	S	-	Test proves a reviewed result is unchanged by polling
L4-03	Background-check gauge on the company with the moves, roles and requirements in 3.3	M	L2-05, L1-11	Every move in the table works; others refused
L4-04	Decisions as locked, superseding records (0015); reopen and reassess by one COMPLIANCE/ADMIN user with a reason; replace the one-review-only rule with superseding reviews	M	L4-03	Reopen works; the original decision is unchanged in the database
L4-05	"Became CLEAR" signal for Dev 2	S	L4-04	M4 passes with L2-11
L4-06	Evidence snapshot: store document and check-result IDs with each decision	S	L4-04, L3-09	Later uploads do not change a past decision's evidence list
L4-07	Honest placeholders: manual PASSED needs evidence; no pending-forever rows; bank panel says "not connected" [E18][E19]	S	-	Items in 5.4 fixed or clearly labelled
L4-08	Risk scale LOW/MEDIUM/HIGH/CRITICAL; CRITICAL shown distinctly [E40]	S	L4-03	Chip and filter show CRITICAL
L4-09	Screening checklist: full history per item; item keys checked by the server [E15]	S	-	Unknown keys refused; history visible
L4-10	RXIL results intake: stored as given (provider rxil), never recomputed; starts the check (A8)	M	L4-03, L2-12	Sample RXIL package fills the check
L4-11	Buyer checks recorded against the buyer, never the company [E35]	S	L3-06	A failed buyer check leaves the company unchanged
L4-12	Handover events: company.became_customer and deal.handed_over, documented for other teams	S	L1-12	Payloads match 3.6; documentation merged
L4-13	Background-check panel: gauge, decisions and history, reopen dialog with reason, checks, screening items	L	L4-03 to L4-09	Usable end to end on sample data
L4-14	End-to-end test of sections 4.1 and 4.2 with Dev 2 and Dev 3	M	All	M5 test passes in CI

Watch out for

- The database makes each review permanent. Keep that: reopening adds a new record, it never edits.
- Do not connect the case engine, Middesk, Trulioo or Sumsb in the prototype (A6, A13).
- The frontend verification panel has no tests today; add them as you rebuild it.

10. Risks and watch items

Risk	Why it matters	What we do	Owner
Company identity takes longer than planned	Search and lists depend on the legacy table; everything hangs off the company	Do L2-03 first; Dev 1 takes screens if Dev 2 slips	Dev 2
File contention	Four files carry most changes	Week 0 split; owners review shared files	Dev 1, Dev 2
Reversible decisions vs locked database	Reviews cannot be changed by design	Superseding records, never edits	Dev 4
Messy import data	Thousands of leads with inconsistent identifiers	Template, validation report, PAN-based matching	Dev 2
RXIL format unknown	Their package shape, matching and document delivery are not specified	One shared parser; treat the format as provisional; isolate mapping in one place	Dev 2, Dev 4
Placeholders mistaken for real results	Demo viewers may trust fake checks	Labels; evidence required; gate in 7.6	Dev 4
Thin frontend tests	Only four frontend test files today	Tests for every new panel	All
Building on legacy code by mistake	Legacy onboarding and the case engine look usable	Section 2.4; code review	All
Customers team never builds the receiver	Payments and compliance will not see new customers	Events are documented and stable; raise at M4	Dev 4

11. Outside our control, and later work

Item	Who	Status
RXIL integration specification: one or many deals per package, files or references for documents, how to match an exporter	RXIL	Awaited. Intake built as provisional.
Receiver for company.became_customer	Customers team	Later (decision 10)
Receiver for deal.handed_over	Lending team	Later
Real providers (Middesk, Trulioo, Sumsub) and credentials	Compliance, platform	After the prototype
Real virus scanner; S3 with Object Lock and KMS	Platform	Before real documents (7.6)
Field-level encryption for tax IDs	Platform	Before production
Connecting the case engine to background-check decisions	CRM	After the prototype (A6)
Automated qualification	CRM	Later; data already records manual or automated
Relationship-manager ownership and unmasked IDs for owners	Business	After the prototype (decision 12)
Documents to correct: blueprint, frontend ticket document and README (still describe the old line and invoices); Jira 4.1-S4T3 (names GCP, now AWS)	Programme lead	Update when convenient

Appendix A. Audit evidence

Paths are relative to backend/app/ unless marked repo: or frontend:.

Key	What it shows
E1	Backend suite on a throwaway local database: 2,797 passed, 22 failed, 5 errors, 6 skipped. All 27 get 404 from payments/FX routes not mounted in <code>api/rest/router.py:1-38</code> .
E2	Frontend <code>tsc -b && vite build</code> : exit 0; 517 kB chunk warning. Vitest: 4 files, 24 tests pass.
E3	<code>frontend:.gitignore:13-14</code> ignores <code>openapi.json</code> and <code>schema.ts</code> ; <code>types.ts:1</code> imports them.
E4	71 migrations, one head (<code>onboarding_0012_risk_critical</code>). Duplicates: <code>compliance_0002</code> (two), <code>rails_0003</code> (two). Next onboarding number 0013.
E5	<code>modules/onboarding/domain/entities/exporter_enums.py:35-53</code> : lifecycle list.
E6	<code>application/exporter_profile_service.py:106-128</code> (13 moves), 134-150 (gated states), 387, 664-668 (enforcement).
E7	<code>frontend:src/modules/onboarding/constants.ts:14-48</code> grouping, 103-117 moves, 126-139 gate.
E8	Frontend users: StageChip, LifecycleMoveControl, ExportersListPage, PipelinePage, AddExporterPage:90, plus three page tests.
E9	18 backend files use the lifecycle, including <code>exporter_router.py</code> , <code>schemas/exporter.py</code> , both repositories, migrations 0005 and 0011, five test files.
E10	Migration <code>onboarding_0005</code> creates profile, contact, activity tables; no foreign keys; only <code>customer_id</code> unique.
E11	<code>exporter_profile.py:115-176</code> has no name column; <code>create_lead</code> (service 236-250) writes the name to <code>OnboardingRequest</code> ; search joins it at 545-549.
E12	Service line 236 generates <code>customer_id</code> randomly; 0 match customers.customers.
E13	22 tables use <code>prevent_mutation()</code> , including <code>screening_review_item</code> and <code>exporter_lifecycle_history</code> (migration 0011); <code>exporter_profile</code> protects only source; <code>verification_result</code> protects <code>reviewed_by</code> and <code>review_status</code> .
E14	History written at service 586-621 (terminal flag 618); <code>list_by_customer</code> and <code>find_transition</code> never called; no route.
E15	<code>screening_review_service.py:43-54</code> eight keys; schema Literal at <code>schemas/exporter.py:422</code> , column <code>String(32)</code> ; insert per save 142-153; latest only 83-90; history method 96 unrouted.
E16	<code>exporter_contact_activity_service.py:42, 88</code> : no company existence check; pending = has due date (139-195).
E17	<code>verification_service.py:299</code> adapter; 418-475 polling overwrites status/risk; 496-537 one-time review (second review 409).
E18	<code>manual_entry_adapter.py:69-147</code> accepts any status; <code>stub_rxil_adapter.py:63,176</code> ; Middesk/Trulioo unreachable from CRM (<code>schemas/verification.py:21</code>); Sumsb off; placeholder generator <code>VerificationSection.tsx:48,556-575</code> .
E19	<code>exporter_router.py:584-598</code> hard-codes <code>connected_accounts=0</code> .
E20	<code>compliance/infrastructure/screening.py:63-84</code> exact-name match; runs only on transactions (<code>compliance/application/services.py:116-131</code>).
E21	cases module has no <code>api/</code> folder; only <code>bootstrap.py:247</code> imports it.
E22	<code>customers/api/router.py:3</code> empty router; nothing writes Customer; entity types BUYER/SUPPLIER/BOTH.
E23	Two <code>ComplianceCase</code> (<code>cases/.../compliance_case.py:42</code> , <code>compliance/.../compliance.py:107</code>); two Customer; two <code>CaseRepository</code> ; onboarding case.py:46.
E24	63 mounted routes; frontend calls 15; CRM routes require a staff role (<code>exporter_router.py:79-91</code>).
E25	<code>api/rest/auth/router.py:37-65</code> forces API_USER; <code>auth/schemas.py:8-19</code> refuses role; no admin bootstrap.
E26	Login-only routes: <code>audit/api/router.py:75,94,119</code> ; <code>compliance/api/router.py:91,110</code> ; <code>api/rest/workflows.py:45,95</code> .
E27	<code>notifications/api/router.py:15-37</code> : no auth dependency.
E28	<code>schemas/verification.py:21</code> ; <code>workflow_dependencies.py:658-697</code> ; <code>kyb/domain/ports.py:60-104</code> . "os.system" refused.
E29	<code>schemas/case.py:108</code> caller chooses source; <code>case_service.py:227-229</code> , 292-307 blanks actor for SYSTEM; <code>tests/integration/test_state_machine_api.py:357-362</code> asserts it.

Key	What it shows
E30	schemas/exporter.py:34-104 masks on the server; DEVELOPER search by full PAN returns the company; relationship_manager_user_id set only by tests.
E31	service 488-519; line 506 drops None values; no name field in the update request.
E32	Duplicate PAN accepted (201); 16 duplicate PAN and 18 GSTIN groups locally.
E33	No CRM service calls AuditService or the event bus; publisher used only by legacy onboarding_service.py:100 and webhook_service.py:140.
E34	onboarding_document.py:27-45 hangs off onboarding_request; no upload, storage or scan code; integrations/object_storage empty.
E35	orchestration_enums.py:111-198 includes INVOICE, BUYER, INVOICE_DUPLICATION; type-pair table verification_service.py:92.
E36	lint-imports: 19 contracts kept; importlinter.ini:296; kyb/domain/ports.py:60-62 names onboarding's adapter package.
E37-E38	Local row counts (mostly test residue); live probes: LEAD → ONBOARDED with nothing, ghost-company writes accepted.
E39	frontend/layout/Sidebar.tsx:18-26 Follow-ups "soon"; DashboardPlaceholder.tsx.
E40	Four risk list types: customers.risk_rating_enum, review_risk_rating_enum, onboarding_risk_rating_enum, verification_risk_level_enum.
E41	domain/policies/state_machine.py:61-138: 12 states incl. MORE_INFO_REQUESTED, REOPENED_BY_EXCEPTION.
E42	Bank-activity route only in test_route_authorization.py; no VerificationSection tests.

Appendix B. Glossary

Term	Meaning
Journey	The company's main stage: LEAD, PROSPECT or CUSTOMER. Moves forward only.
Gauge	A separate status on the company answering one question: qualification, conversation, background check.
Deal	A specific current financing need of a company (sometimes called an opportunity). Many per company.
Marker	ENDED or PAUSED flag with a reason, for commercial endings or pauses.
Qualification	Checking a lead against our business requirements (revenue, age, export history and so on).
Screening checklist	The existing eight-item compliance list inside the background check. Not qualification.
Background check	Compliance checks on the company: KYB, KYC, AML, sanctions and review.
KYB / KYC	Know Your Business / Know Your Customer: verifying who a company and its people are.
AML / CFT	Anti-money-laundering / countering the financing of terrorism.
PAN, GSTIN, IEC, CIN	Indian identifiers: tax account number (one per company); GST registration (one per state, contains the PAN); importer-exporter code; company registration number.
RXIL	Partner that sends pre-filtered exporters with its own check results.
Locked table (append-only)	Rows can be added but never changed or deleted, enforced by the database.
Superseding record	A new decision that replaces an earlier one without editing it; both stay visible.
Evidence snapshot	The list of documents and results a decision relied on, stored with the decision.
Event	A message the CRM announces (for example "became customer") for other systems to act on.
Contract	A short written agreement between developers about a shared data shape or rule.
Migration	A numbered, versioned change to the database structure.
Legacy onboarding path	The old Temporal-based workflow. Out of scope; never build on it.

Term	Meaning
Module rule	Connecting code lives in onboarding; kyb, cases, customers and compliance are not edited.
Roles	OPERATIONS (sales and relationship staff), COMPLIANCE, ADMIN, DEVELOPER (read-only), API_USER (lowest).

Appendix C. Architecture poster

The next and final page is the full architecture poster as a large vector page. Zoom in to read it; it prints best on A3 or larger. The same poster is available separately as a PNG.

One company, several gauges

How the new CRM follows a company: one short main journey, plus separate gauges that each move on their own.

